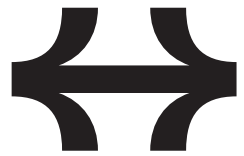


The hidden costs of AI

The environmental footprint of AI and the data centres that power it





**Every time you use AI,
it uses electricity and
water. Should we care?**

How much should we care?

- 1 — Not at all
- 2
- 3
- 4
- 5 — Enormously

What's under one question

1%

Electricity

of global electricity, projected to double by 2026

5 cars

Per model trained

of CO₂ over their entire lifetimes

Nature Sustainability 2025 · International Energy Agency 2025 · MIT

32.6–79.7 Mt

Carbon

AI's estimated 2025 footprint — about New York City's

≈ bottled water

Water for cooling

projected 2025 use, about the whole bottled water industry

So — should we care?

A

Yes — it adds up

Small individual actions combine into a large collective impact.

B

This is not "never use AI"

It is "use it thoughtfully".

C

Companies have a job too

Transparency, efficiency and renewable energy.

D

And so do we

We can choose when AI is worth it, and ask for sustainable AI.

When is it worth it?

Rarely worth the cost

A question you could answer yourself

10

a search would do

Jokes and trivial entertainment

24

Clearly worth the cost

Work you then rewrite yourself

52

Accessibility needs

82

Medical research, climate modelling

94

Categories and ordering from the AI Awareness Day teacher pack

Two words worth knowing

Data centre

A building full of servers. AI needs large ones, using electricity for computing and water for cooling.

Carbon footprint

The total greenhouse gases caused by an activity: for AI, electricity, hardware and cooling.

Using AI responsibly

01 **Think first**
Do you really need AI for this task?

02 **Check yourself first**
Simple questions often don't need AI at all.

03 **Ask for transparency**
We need to know AI's true environmental cost.

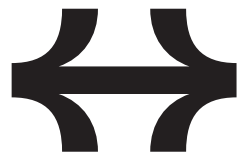
Three habits — none of which is "stop"

Keep humans in the loop



Thoughtfully — not never

Every choice we make about technology has consequences. Use AI thoughtfully.



**Tech companies, not users,
should be responsible for cutting
AI's environmental impact.**

Where do you stand?

- For
- Against
- Not sure yet

Build both sides

For	Against
Companies design the models and run the data centres, so they control most of the impact.	Companies respond to demand, so users' choices shape what they build.
Users cannot see how much energy or water a request uses, so they cannot make informed choices.	Governments set the rules on energy and water, so responsibility is shared.
Companies have the money and expertise to switch to renewable energy and better cooling.	Every user can choose when AI is really needed.

Challenge card: What if companies only change when their customers demand it?

Keep humans in the loop

The same debate, at every age

A

Primary · Years 5 and 6

Should we use AI less to help look after the planet?

B

Secondary · Years 7 to 11

Tech companies, not users, should be responsible for cutting AI's environmental impact.

C

Post-16

This house believes the benefits of AI are worth its environmental cost.

If any of this affected you

You will not be in trouble for asking for help.



Reporting and support

Childline — 0800 1111

<https://www.childline.org.uk>

Samaritans — 116 123

<https://www.samaritans.org>

Young Minds — mental health

<https://www.youngminds.org.uk>

CEOP — report abuse or exploitation

<https://www.ceop.police.uk/safety-centre>

Internet Watch Foundation — report an image

<https://report.iwf.org.uk>

UK Safer Internet Centre

<https://saferinternet.org.uk>